

Sanity Checks for Reinforcement Learning Explanations

Abstract. We show that a feature attribution for a reinforcement learning agent can be stable across training runs, precisely estimated, semantically plausible, and carry no information. On an empirically efficient prediction market, a PPO agent whose return is statistically indistinguishable from abstention produces attributions with cross-seed rank correlation 0.849 and unanimous agreement on the most important feature. We give a null-model test, built on the additivity identity attribution schemes already satisfy, which compares an agent’s attribution span against agents trained where the observation channel carries no information. It rejects a planted signal at $z = 12.27$ and fails to reject the market agent at $z = +0.23$. The established sanity check for this purpose, randomising network parameters, does not transfer: its reference distribution is $42\times$ wider on CartPole, and across eight checkpoints on two control tasks it never exceeds $z = 2.45$, including on a converged policy. Finally, the market agent’s attribution can be driven from 39.9% to 3.2% of total mass at no measurable cost in return, while the same intervention on a feature that genuinely carries signal destroys 81% of return.

1 Introduction

Interpretability is proposed as a basis for oversight: if we can see which inputs drive a policy, we can decide whether to trust it. This requires that an attribution mean something, and attribution methods return an answer regardless. Given any policy and state, Shapley values [6, 9] and integrated gradients [11] produce a ranked, signed vector. Nothing in the output separates an attribution reflecting structure the agent exploits from one reflecting nothing.

Supervised learning handles this with randomisation tests. Adebayo et al. [1] compare explanations of a trained model against a model with randomised parameters, and against a model trained on permuted labels. Reinforcement learning has adopted only the first. Huber et al. [5] apply the parameter test to Atari saliency and observe that the data-randomisation test has no obvious RL analogue: there is no labelled dataset to permute.

We supply that analogue, and show the half already adopted does not work.

Contributions.

1. An environment-level null for RL: corrupt the observation channel, leaving dynamics and reward intact, so the agent trains normally with nothing to condition on. No access to task internals is required.
2. Evidence that parameter randomisation has near-zero power in RL.
3. A ground-truth validation protocol using environments where the correct attribution is known by construction.
4. A demonstration that attribution is not identified by behaviour.

What we do not claim. The argument that RL attribution should target outcomes rather than policy outputs is due to Beechey et al. [3]; FastSVERL [2] addresses the scalability, off-policy and temporal prob-

lems we only name. Our test adapts Adebayo et al. [1]. Deletion-based fidelity is established practice in XRL [4, 7]. Our contribution is the RL construction, the demonstration that the incumbent fails, and the consequences.

2 Method

Let $v(S)$ be the expected episode return of an agent observing only the features in $S \subseteq N$, with features outside S replaced at every timestep by draws from a reference distribution. Define the *attribution span* $\Delta = v(N) - v(\emptyset)$.

Shapley values and integrated gradients satisfy additivity identities under which per-feature attributions sum to Δ , so the span can be read off either. It is worth stating that Δ is *not* a Shapley quantity: it is the difference between two masked rollouts and depends only on the masking. This has a practical consequence: it costs two evaluations rather than $2^{|N|}$.

Blind-observation null. *Replace every observation with an independent draw from a fixed distribution matched to the environment’s observation moments. Leave dynamics and reward untouched. Train normally.*

Such an agent faces the real task and receives real return but cannot condition on anything. Unlike parameter randomisation it yields a *normally trained* policy: the failure mode that occurs in deployment. Matching the reference moments matters, or the network fails for reasons of scale rather than information.

Given null draws with mean $\bar{\Delta}$ and standard deviation s we report a rank p -value and $z = (\Delta_{\text{obs}} - \bar{\Delta})/s$, and require both to reject. The rank statistic assumes no shape but bottoms out at $1/(n+1)$; the normal statistic has resolution but assumes a shape the null need not have. If every null draw is identical, $s = 0$ and any deviation is maximally surprising: we return $z = \pm\infty$ and surface the degeneracy rather than guarding it away.

Table 1: Test split, 1,284 held-out episodes. Paired bootstrap against always-flat.

Policy	Mean P&L	Trades/ep	p vs flat
always-flat	+0.000	0.00	n/a
buy-and-hold	-1.608	1.00	0.23
PPO (5 seeds)	-0.595 ± 0.176	1.37	0.10
mean-reversion	-11.615	4.94	<0.0001
random	-18.532	9.33	<0.0001

3 Setup

Environments with known answers. Explanation evaluation lacks ground truth [4]. We construct fixed-length synthetic corpora where the answer is fixed by construction: a *planted signal* (one of 18 features informative about the outcome), a *null* (none), and a *decoy* (one feature correlating 0.989 with the outcome in-sample and 0.006 out of sample).

Market. Our main setting is a binary contract settling to \$1.00 if BTC is higher after 15 minutes, on a regulated prediction market: 6,425 settled contracts over 68 days, walk-forward split by time, test evaluated once. Every episode resolves to a known value. Agents are trained with PPO [8]; reward is the change in mark-to-market equity net of the exchange’s published fee schedule, a 9% round-trip hurdle at the price this contract trades at.

Control tasks. CartPole-v1 and Acrobot-v1, whose observation spaces are small enough to enumerate all coalitions, so Shapley values there are exact.

4 Results

4.1 An explanation that passes every check

The market is empirically efficient: its implied probability has a weighted mean absolute calibration error of 0.0172, and spot return correlates 0.476 with the outcome but only 0.016 with the residual after the price is accounted for. No policy beats abstention (Table 1).

PPO’s shortfall is its own: on synthetic data with no signal by construction the same agent scores -0.47 ± 0.47 . Yet across five seeds whose pairwise performance differences are indistinguishable in all ten pairs, its attributions have cross-seed rank correlation 0.849 with 100% agreement on the top feature, a top-5 ranking whose adjacent gaps exceed their combined standard errors, and a legible story: time-to-expiry dominates, reading as an agent that learned when not to trade.

The null test disagrees (Table 2, Figure 1). It has power and specificity, and the market agent sits at the centre of the distribution of explanations of nothing.

We note one consequence for our own reporting. An earlier draft stated that “the whole observation is worth +5.914 per episode against being blind” and presented it as a finding. Blind agents produce $+8.19 \pm 5.11$. It was noise.

Table 2: Null-model test. Null from 24 blind-trained agents, span $+8.19 \pm 5.11$.

Case	Span	z	Verdict
planted signal	+70.89	+12.27	informative
null corpus	+11.86	+0.72	not distinguishable
real market	+9.37	+0.23	not distinguishable

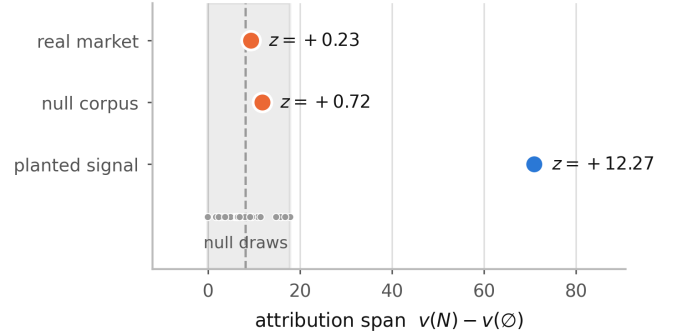


Figure 1: Attribution spans against the null. The band is the range over 24 agents trained where the observation channel carries no information.

4.2 Validation against ground truth

On the planted corpus, outcome attribution ranks the planted feature first with 55% of total attribution mass. On the decoy corpus, an agent trained where the decoy predicts the outcome (in-sample +47.24, held out -6.83) is explained on episodes where the decoy is provably worthless: per-decision attribution gives it 6.5% of mass at rank 3 of 18, outcome-based attribution 1.1% at rank 10. Both statements are true; the policy does key on the decoy. Per-decision attribution is correct about behaviour and misleading about value.

4.3 The established null does not transfer

We build both nulls for the same task (Table 3, Figure 2). A randomly initialised policy behaves arbitrarily, so masking its inputs moves return unpredictably and the reference distribution is very wide. Across eight checkpoints the parameter null never exceeds $z = 2.45$, including on a converged CartPole policy scoring 404 of 500, while the environment null reaches $z = +117.6$. The two disagree on five of eight.

Acrobot also shows the test tracking *whether there is anything to explain* rather than agent quality: its agent fails completely until it learns, the span is exactly 0.00 there, and the test correctly declines. CartPole’s observations matter at every skill level and the test fires throughout. We do not claim undertrained agents generally produce empty explanations.

4.4 Attribution is not identified by behaviour

We add a training penalty on the divergence between $\pi(\cdot | s)$ and $\pi(\cdot | s')$, where s' has one feature resampled from the batch marginal: the same perturbation the attribution measures (Table 4, Figure 3).

The attribution is equally suppressible in both cases.

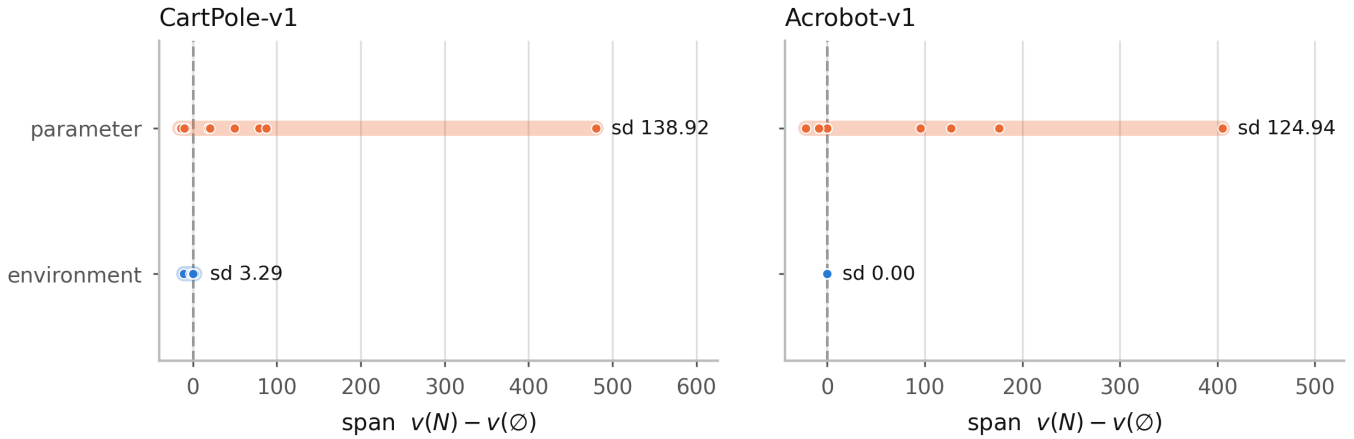


Figure 2: Spread of each null construction. Width determines power: a very wide reference distribution cannot reject anything.

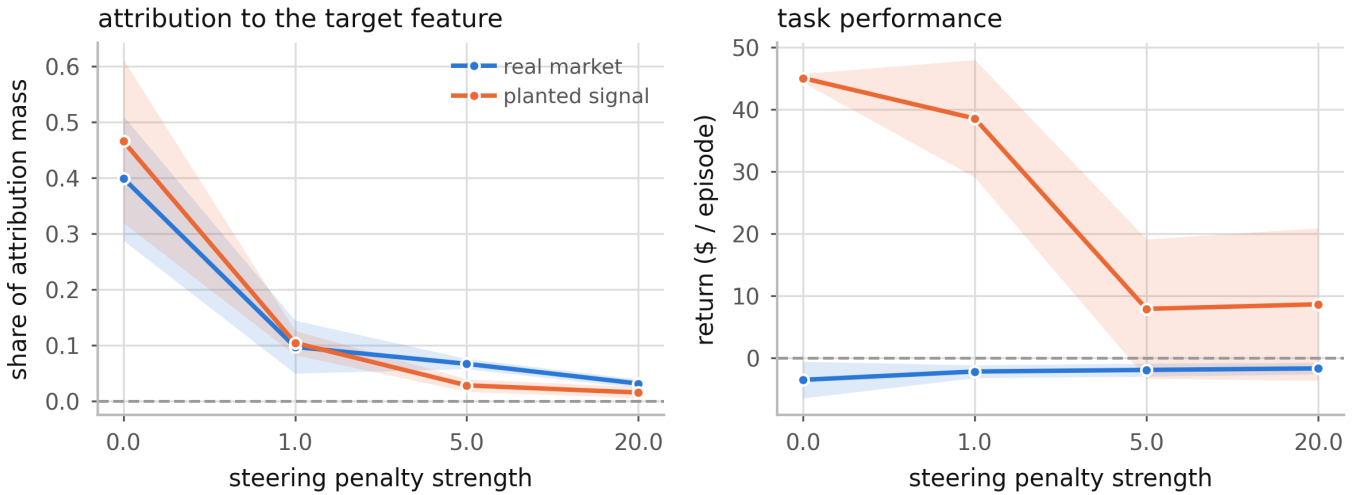


Figure 3: Attribution is equally suppressible on both corpora (left). Only where the feature is load-bearing does suppressing it cost return (right).

Table 3: Null construction on two control tasks.

Environment	Parameter	Environment	Ratio
CartPole-v1	± 138.92	± 3.29	$42\times$
Acrobot-v1	± 124.94	± 0.00	degenerate

Table 4: Steering one feature’s attribution.

Corpus	Attribution	Return	p
market	39.9% $\rightarrow$ 3.2%	$-3.49 \rightarrow -1.65$	0.66
planted signal	46.5% $\rightarrow$ 1.5%	$+45.04 \rightarrow +7.92$	<0.001

What differs is the price: suppressing the planted signal destroys 81% of return; on the market it costs nothing measurable. Explanations are steerable at no cost exactly where they are not tracking anything, corroborating Section 4.1 by a route sharing no machinery with the null test. The control matters: had steering succeeded on both, the penalty would merely be defeating the attribution method.

This differs from Slack et al. [10], who construct classifiers that detect the off-manifold points LIME and SHAP query. Here the agent is trained normally and the resulting explanation is *correct* about the resulting model; the point is that behaviour does not pin it down.

5 Limitations

The test adapts Adebayo et al. [1]; the outcomes framing is Beechey et al. [3]’s. The span measures information in aggregate, not whether the decomposition among features is right. Temporal credit assignment and off-policy attribution are named, not solved; these are FastSVERL’s [2] territory. We proposed a mechanism for the $42\times$ gap (randomising weights perturbs the measurement’s domain and compounds along the trajectory) and tested it: over horizons 4 to 28 both nulls grow at indistinguishable rates ($\alpha = 0.94$ against 0.83). The hypothesis is rejected and the gap remains unexplained. Scope is one real domain and two control tasks.

6 Conclusion

An attribution can be stable, precisely estimated, plausible, and empty. The checks practitioners apply do not detect this, because none of them ask whether the explained agent learned anything. A null-model test does, using a statistic the attribution framework already provides. The established test for this purpose does not survive the move to reinforcement learning.

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